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RÉPUBLIQUE FRANÇAISE

MINISTÈRE DE LA DÉFENSE



ICACM 2008

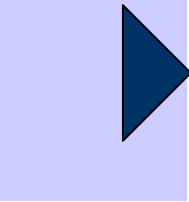
Organization meeting

21 September 2007



DÉLÉGATION GÉNÉRALE POUR L'ARMEMENT





Plan

- Context
- Objectives et characteristics of ICACM2008
- Rocamadour & its region
- Topics, committees, invitations status
- Organization, costs, funding
- Web site
- Poster
- Questions

Context (1/2)

- Objective: promote research partnership between US & FR universities in applied & computational mechanics
- Initiative: REEF (Research & Engineering Education Facility), University of Florida, 2005
- ICACM = International Center for Applied Computational Mechanics
 - members = REEF (Univ. Florida), Polytech Lille, INPG, Paris XIII, Univ. Savoie, Univ. Michigan
- Objective of an annual symposium thématique annuel, alternatively in France & USA
 - 2007: Bridging Scales in Computation (REEF, Florida, USA)
 - 2008: Materials under extreme loadings (Rocamadour, France)
 - 2009: Fluid mechanics (USA)
 - 2010: Energetics (FR)
- Visit of REEF to FR in February et June 2006
 - Visit to CEG, 2006, 20th June





Context (2/2)

- First event:
Advances in Bridging Scales in Computation
From Microstructure to Macro-Scale Properties of Heterogeneous Materials
March 28-30, 2007 - Shalimar, FL
- Funding: AFOSR (Air Force Research Laboratory)
- 40 participants, 2/3 French
- 28 presentations, 12 keynote speakers, 10 US presentations
- 90% of participants were speakers:
 - Keynote speakers: 45 minutes
 - Invited speakers: 30 minutes
- Registration, lunches, lodging: free



Objectives of ICACM2008

- Title:

MATERIALS UNDER EXTREME LOADINGS - APPLICATION TO
PENETRATION AND IMPACT

May, 28-30, 2008, Rocamadour, Hôtel du Château, France

- Objective of 60 participants, 2/3 US, under invitation solely
- DGA: Political support from D(CEG), Métier MA, Pôle MC
- Funding: DGA/UM-MID + CNRS + DYMAT
- Contract n°33062 (lot 2, task P1), PEA CODYMA n°070706
- Presentations to the ICACM executive advisory board committee:
 - 30 March 2007
 - 21 September 2007

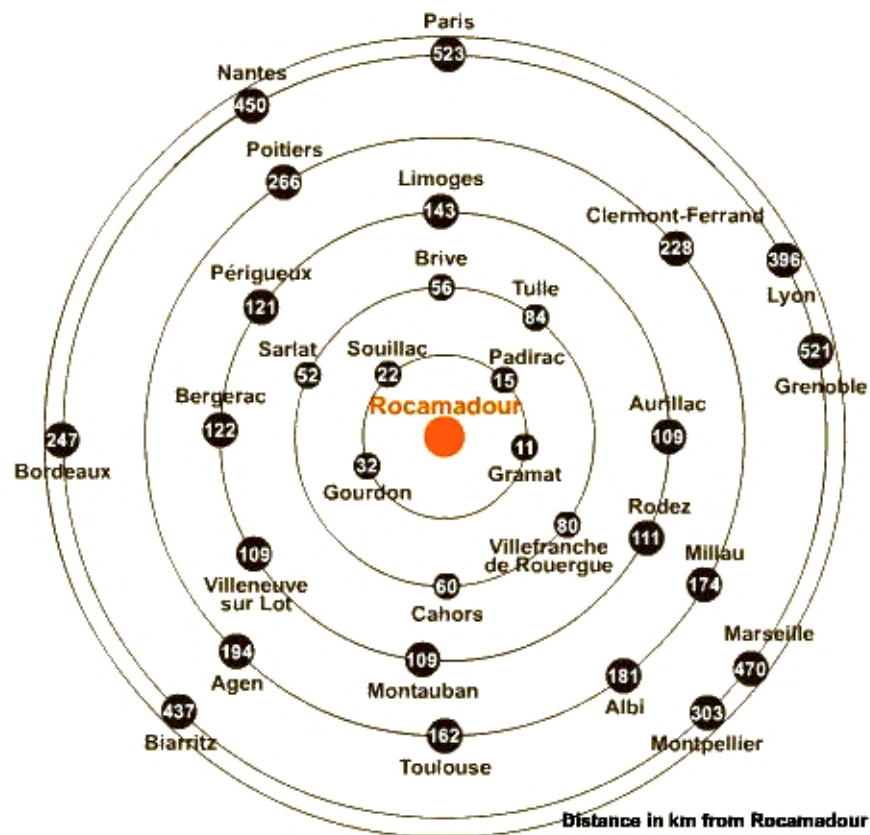
Conference location



The Lot, one of the 9 counties of the Midi Pyrénées region

How to get there

- By road:
 - N140 (Brive-Rodez) via Gramat
 - A20 (Paris-Toulouse): exit 54, 55 or 56
- By train:
 - The train can take you to Rocamadour station by direct journey. Approx. 5:10 to come from Paris (Paris-Rodez line)
- By Air:
 - Airport of Brive (57 km)
 - Airport of Rodez-Marcillac (100 km)
 - Airport of Perigeux-Bassilac (121 km)
 - Airport of Limoges (142 km)
 - Airport of Toulouse (162 km)



Conference location

- Parc naturel régional des Causses du Quercy:
 - created in 1999
 - 96 villages & towns
 - 176 000 hectares
- Caussenarde geography:
 - Sheeps on the plateau
 - Men in the valleys
 - Water inderground



Quercy

Nothing will stop me believing that if Cro-Magnon man moved here, it was because he was deeply intelligent with a very developed sense of beauty. Nothing will stop me believing that religious feeling had already reached a high degree of development in him and that it bloomed in these places, when that same man lived like a beast at the bottom of caves. Nothing will stop me believing that this great and peaceful region of France is destined to stay a sacred place eternally for man and that, when the big city has finished wiping out poets, their successors will find shelter and support here.

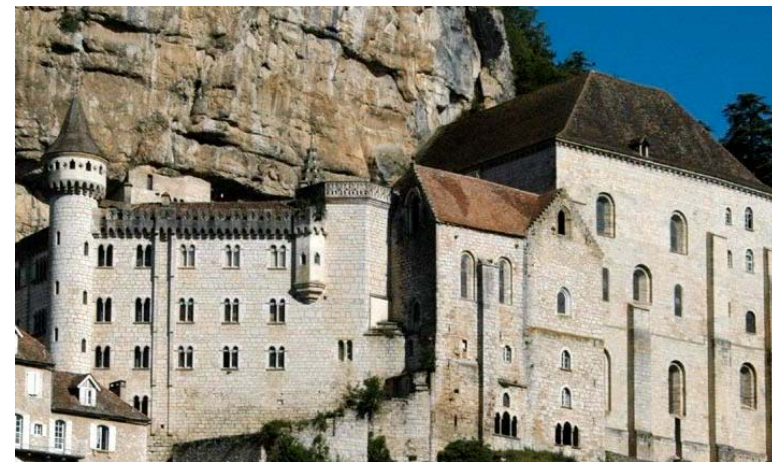
This visit to the Dordogne was for me, I repeat, of a fundamental importance: it remains with me as a hope for the future of the species, and even of our planet. It may be that one day France will stop existing, but the Dordogne will survive, just like the dreams that feed the human soul."

Henry Miller - Excerpt of *"The Colossus of Maroussi"*

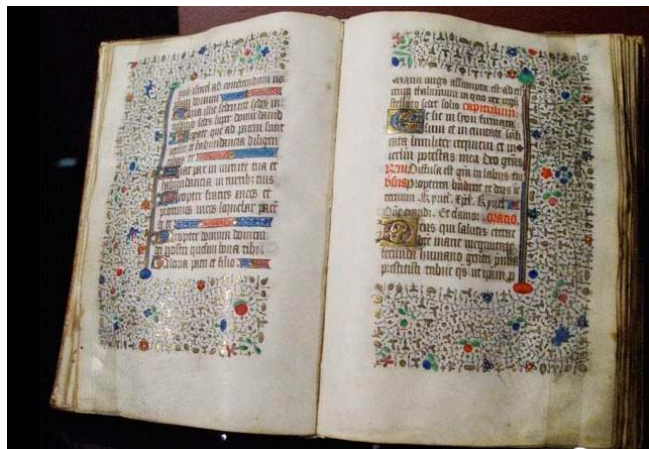


Rocamadour

- One of the most famous pilgrimage places in Europe since XIIIth century
- Gorge of 150 m depth cut by the Alzou river
- 632 inhabitants
- More than 1 500 000 visitors per year
- 126 miracles assigned to virgin Mary (1172)
- The Great Stairway of 216 steps



Rocamadour



The houses on the rock , the churches on the houses, the rock on the churches, the castle on the rock (Pierre Loti)



ICACM08 topics

Topics :

- Characterization of material behaviour under extreme loading (damage, moisture content, influence of pressure,...)
- Microscopic observations, analysis of damage & mesoscale modelling
- Macroscopic modelling, damage & plasticity
- Penetration mechanics, trajectory instabilities
- Localization, cracking, dynamic fragmentation of materials, modelling & simulation

Committees

Committees :

Scientific committee:

Pr T. Belytschko, USA, Northwestern University
Pr D. Bresch, France, LAMA, Université de Savoie
Pr O. Cazacu, USA, REEF, University of Florida
Pr F. Darve, France, L3S-R, Université Joseph Fourier
Dr R. Dormeival, France, CEA, DYMAT
Pr P. Franciosi, France, Université Paris XIII
Pr D. Kondo, France, Polytech'Lille
Dr M. Perez, France, DGA, Centre d'Etudes de Gramat
Pr J. Mazars, France, L3S-R, INPG
Pr A. Molinari, France, LPMM, Université de Metz
Pr. W. Shyy, USA, University of Michigan

Local organization committee:

Dr. E. Buzaud, France, DGA, Centre d'Etudes de Gramat

Invitation status: Keynote speakers

<i>N°</i>	<i>Status</i>	<i>Name</i>	<i>Nationality</i>	<i>Establishment</i>
K1	invited	Dr S. Akers	United States	ERDEC
K2	invited/accepted	Pr R.C. Batra	United States	Virginia Tech
K3	invited	Pr T Belytschko	USA	Northwestern University
K4	invited/accepted	Pr W. Chen	United States	Purdue University
K5	invited/accepted	Pr A. Combescure	France	Insa Lyon
K6	invited/accepted	Pr L. Daudeville	France	L3S
K7	invited/accepted	Dr M.J. Forrestal	United States	Sandia Nat Lab (retired)
K8	invited	Dr D. Grady	USA	ARA co.
K9	invited/accepted	Pr F. Hild	France	ENS/LMT Cachan
K10	invited/accepted	Dr I. Ionescu	France	LPMTM
K11	invited/accepted	Pr A Molinari	LPMM	France
K12	invited/non	Dr Ortiz	United States	CalTech
K13	invited/accepted	Pr William Proud	United Kingdom	Cambridge University
K14		Dr G.R. Johnson	United States	AHPCRC
K15		Pr K.T. Ramesh	United States	Johns Hopkins University
K16		L.E. Schwer	United States	Schwer Consult.
K17		Mr S. Silling	United States	Sandia Nat Lab
K18		Dr J Weerheijm	Netherlands	TNO

K(USA) = 3

K (UK)= 1

K(FR) = 5

US keynote speakers missing!

Invitation status: Invited speakers

<i>N°</i>	<i>Status</i>	<i>Name</i>	<i>Establishment & Nationality</i>
I1	invited/accepted	Mr M.D. Adley	ERDEC (USA)
I2		Pr S Bless	IAT (USA)
I3		Mr P Church	QinetiQ (UK)
I4		Dr G Cusatis	Rensselaer Polytech. Inst. (USA)
I5	invited/accepted	Dr K.T. Danielson	AHPCRC (FR)
I6	invited/accepted	Dr P Forquin	Université de Metz (FR)
I7		Mr A.F. Fossum	Sandia Nat Lab. (USA)
I8	invited/accepted	Mr A.O. Franck	ERDC (USA)
I9		Pr S. Gattu	University of Florida (USA)
I10		Dr J.K. Gran	SRI (USA)
I11		Dr G.R. Johnson	AHPCRC (USA)
I12		Pr Landis	University of Maine (USA)
I13		Pr D. Littlefield	University of Alabama (USA)
I14		Dr C. Mariotti	CEA (FR)
I15	invited/accepted	Dr J. Petit	DGA (FR)
I16		Pr K.T. Ramesh	Johns Hopkins University (USA)
I17	invited/accepted	Mr A. Rouquand	DGA (FR)
I18		Dr A. Rusinek	Université de Metz (FR)
I19		Dr L.E. Schwer	Schwer Consult.(USA)
I20		Mr S. Silling	Sandia Nat Lab. (USA)
I21		Dr G Solomos	JRC (IT)
I22		Pr A. Vaziri	Harvard University (USA)
I23		Pr J. Walton	Texas A&M (USA)
I24		Dr J Weerheijm	TNO, (NL)
I25		Pr T Wierzbicki	MIT (USA)
I26		Pr Worswick	University of Waterloo (USA)

Invitation status: invited Participants

<i>N°</i>	<i>Status</i>	<i>Name</i>	<i>Establishment & Nationality</i>
P1	invited/accepted	Mr B. Patterson	USAF (USA)
P2		Mr T Brantley	USAF (USA)
P3		Mrs D. Chauvel	EDF (FR)
P4		Mr M. Green	USAF (USA)
P5		Mr B. Mortaigne	DGA/CEG (FR)
P6		Mr A. Nachman	AFORS (USA)
P7	invited/accepted	Dr J. Rogacki	REEF (USA)
P8		Dr M. Schmidt	USAF (USA)
P9		Mrs S. Seignolles	DGA/CEG (FR)
P10		Dr Jim Sheridan	DSTL (UK)
P11		MR J.M. Sibeaud	DGA/CEG (FR)
S1		Pr T. Belytschko	Northwestern University (USA)
S2		Pr D. Bresch	Université de Savoie (FR)
S3		Pr O. Cazacu	University of Florida (USA)
S4		Pr F. Darve	Université Joseph Fourier (FR)
S5		Dr R. Dormeval	DYMAT (FR)
S6		Pr P. Franciosi	Université Paris XIII (FR)
S7		Pr D. Kondo	Polytech'Lille (FR)
S8		Dr M. Perez	DGA (FR)
S9		Pr J. Mazars	INPG (FR)
S10		Pr A. Molinari	Université de Metz (USA)
S11		Pr. W. Shyy	University of Michigan (USA)

Accommodations

- Hotel: Hôtel du Château, Rocamadour (www.hotelchateaurocamadour.com)
 - 58 rooms
 - Wifi, tennis court, warmed & covered swimming pool, famous restaurant
 - Conference room, 60 seats, audio/video equipment





Costs

- Hotel: 30000 € TTC
 - 27 May reception diner
 - 28, 29, 30 May light lunches
 - 29 May conference banquet
 - 28, 29, 30 May morning breaks
 - 28 & 30 May afternoon breaks
 - 27, 28, 29 May lodging & breakfast
 - 28, 29, 30 May Conference room



Costs

• Conference banquet animation	1000 €
• Rocamadour visit on the 28 May evening	300 €
• 28 May dinner at L 'Hospitalet restaurant	2000 €
• National transportations	4000 €
• 2 bus from Toulouse Airport to Rocamadour	
• Connection for CEG visit on 29 May afternoon	
• International transportations	10000 €
• 2 flights USA/FR	
• Poster & invitations	1500 €
Total	50 000 €

Poster

**MATERIALS UNDER EXTREME LOADINGS
APPLICATION TO PENETRATION AND IMPACT**



ICACM 2008



THE SECOND FR-US CONFERENCE ORGANIZED BY :

THE INTERNATIONAL CENTER FOR APPLIED COMPUTATIONAL MECHANICS

MAY 28-30, 2008, ROCAMADOUR, HÔTEL DU CHÂTEAU, FRANCE

Scientific committee :

Pr T. Belytschko	Northwestern University (USA)
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Pr O. Cazacu	University of Florida (USA)
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Pr J. Mazars	INPG (FR)
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Keynote speakers :

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Pr R.C. Batra	Virginia Tech. (USA)
Pr T. Belytschko	Northwestern University (USA)
Pr W. Chen	Purdue University (USA)
Pr A. Combescure	INSA Lyon (FR)
Pr L. Daudeville	Université Joseph Fourier (FR)
Dr M.J. Forrestal	Sandia Nat Lab. (USA)
Dr D. Grady	ARA (USA)
Pr F. Hild	ENS Cachan (FR)
Pr I. Ionescu	Université Paris XIII (FR)
Pr W. Proud	Cambridge University (USA)

Distinguished speakers :

M.D. Adley	ERDEC (USA)
Pr S. Bless	IAT (USA)
P. Church	QinetiQ (UK)
Dr G. Cusatis	Rensselaer Polytech. Inst. (USA)
Dr K.T. Danielson	AHPCRC (FR)
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Pr D. Littlefield	University of Alabama (USA)
Dr C. Mariotti	CEA (FR)
Dr J. Petit	DGA (FR)
Pr K.T. Ramesh	Johns Hopkins University (USA)
A. Rouquand	DGA (FR)
Dr A. Rusinek	Université de Metz (FR)
Dr L.E. Schwer	Schwer Consult (USA)
Mr S. Silling	Sandia Nat Lab. (USA)
Dr G. Solomos	JRC (IT)
Pr A. Vaziri	Harvard University (USA)
Pr J. Walton	Texas A&M (USA)
Dr J. Weerheijm	TNO, (NL)
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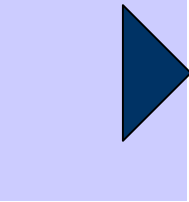
Contact : icacm2008.ceg@det.dga.defense.gouv.fr





WEBSITE

- Presentation of a first layout
- Web site at the University Joseph Fourier
- Email adress: icacm2008@det.dga.defense.gouv.fr
- Reception of inscriptions, abstracts, collect of specific requests



QUESTIONS?